DELFT UNIVERSITY OF TECHNOLOGY
Faculty of Electrical Engineering, Mathematics and Computer Science

CSE2315 Lab Course
Solutions 1

deadline: February 12, 2026, 13:45

1. Consider a language $L = \{ok, a, bad, dab, abba, hi, \varepsilon, acc, duck\}$.

   (a) Give a possible Σ such at $L \subseteq \Sigma^*$.

   > **Solution:** $\Sigma = \{a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z\}$ (Or any other superset of $\{o, k, a, b, d, h, i, c, k\}$

   (b) Why is this only possible Σ?

   > **Solution:** Any superset will do.

   (c) Give all words in L in "modified lexographical order" (or shortlex order). (Note that we often refer to this just as a lexographical order.)

   > **Solution:** $\varepsilon, a, hi, ok, acc, bad, dab, abba, duck.$

2. *Old lab question that was also used on the resit of 2021-2022 (modified):* Consider the following claims (a) and (b). For each claim, verify whether it is true for arbitrary languages $L_1 \subseteq \Sigma_1^*,\ \ L_2 \subseteq \Sigma_2^*,\ \ L_3 \subseteq \Sigma_3^*,\ \ L_4 \subseteq \Sigma_4^*$. If a claim is true, give a proof; if it is not true, give a counterexample with an explanation how the counterexample shows the claim is false.

   (a) If $L_1 \cup L_2 = L_3 \cup L_4$, then $\Sigma_1 \cup \Sigma_2 = \Sigma_3 \cup \Sigma_4$.

   > **Solution:** This claim is false. A counterexample is given by $L_1 = L_2 = L_3 = L_4 = \{aa\}$ and $\Sigma_1 = \Sigma_2 = \{a\}, \Sigma_3 = \Sigma_4 = \{a, b\}$. Then $L_1, \ldots, L_4$ are indeed languages over their alphabets, and also $L_1 \cup L_2 = \{aa\} = l_3 \cup L_4$. But $\Sigma_1 \cup \Sigma_2 = \{a\} \neq \{a, b\} = \Sigma_3 \cup \Sigma_4$.

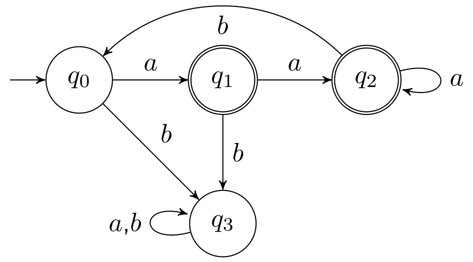
   (b) If $L_1 \cup L_2 \subseteq L_3 \cap L_4$, then $L_1 L_2 \subseteq L_3 L_4$.

   > **Solution:** This claim is true. Consider an arbitrary word $w \in L_1 L_2$. Then w can be written as uv with $u \in L_1, v \in L_2$. So also $u \in L_1 \cup L_2$, and the assumption implies $u \in L_3 \cap L_4$. In particular, $u \in L_3$. Similarly, $v \in L_2$ implies $v \in L_1 \cup L_2$, so $v \in L_3 \cap L_4$, so $v \in L_4$. Combining these two facts, we get $w = uv \in L_3 L_4$.

3. Suppose we have an alphabet $\Sigma = \{a, b\}$. Construct a DFA $M = (Q, \Sigma, \delta, q_0, F)$ that recognizes the following language $L \subseteq \Sigma^*$:

   $$L = \{w \mid \text{each } b \text{ in } w \text{ is immediately preceeded by at least two } a\text{s and } w \text{ ends with an } a\}.$$

   (a) Give the transition graph of M. Use no more than 6 states.

   > **Solution:** The following transition graph gives a DFA that recognizes L:
   >
   > 
   >

(b) Describe briefly how M works.

Solution: States q_1 and q_2 are accepting states, both of which represent that the last symbol read was an a. Additionally, these states represent that we have read 1 a and 2 or more a's since the last b respectively. In the latter case, if we read a b, the last symbol is not an a anymore, and we should reset the counting of a's, wich is handled by the transition $q_2 \to q_0$. If we read a b when we haven't read 2 or more a's, then we go to a trap state q_3, because the word will never satisfy the predicate of L any more.

If we read an a in the start state q_0, the word (i.e., a) is in L, so we go to q_1. If we read an b in q_0, we go to q_3 just like if we read a b in q_1.

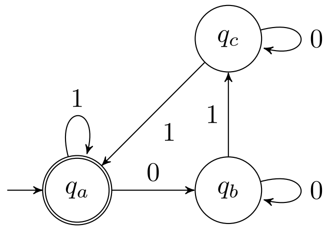
4. *Old exam question (modified), midterm 21-22* Consider the following DFA $D = (\{q_a, q_b, q_c\}, \{0, 1\}, \delta, q_a, \{q_a\})$

where δ is represented using the following table:

	0	1
q_a	q_b	q_a
q_b	q_b	q_c
q_c	q_c	q_a

(a) Give a transition diagram that corresponds to D.

Solution:



(b) Give a word w of length 3 that is not in D but is in $D' = (Q, \Sigma, \delta, q_0, F \cup \{q_c\})$. Explain your answer by giving the state transitions for w.

Solution: Pick any word of length 3 that ends in q_c: 101, 001 or 010. These all transition to q_c via q_b, taking 1 self-loop on the way.

(c) Now imagine that we create a DFA $D'' = (Q \cup Q_2, \Sigma, \delta_1, q_a, F)$, with

$\delta_1(q, s) = \begin{cases} \delta(q, s) & \text{if } q \in Q \\ \delta_2(q, s) & \text{else} \end{cases}$ for some function $\delta_2 : Q_2 \times \Sigma \to Q \cup Q_2$

(Subquestion not included on exam for time reasons): What do we know about δ_2, specifically how many extra transitions will it have to contain? You may assume $Q \cap Q_2 = \emptyset$. Explain your answer.

Solution: For every state there is exactly one transition for every symbol of the alphabet. Hence it grows by $|\Sigma| \cdot |Q_2|$ (every new state will have $|\Sigma|$ transitions).

(d) Describe as accurately as possible the relation between $L(D)$ and $L(D'')$ in terms of $\subset, \subseteq, =$ relations. Explain your answer.

Solution: $L(D) = L(D'')$ as every word that D accepts will take the same path in D'' since $\delta_1^* = \delta^*$ for $w \in \Sigma*$ and as F did not change the word will still be accepted. Furthermore all the new states are thus unreachable as no existing transitions are modified. Hence no new words can be accepted and so $L(D) = L(D'')$.

5. *Old exam question (modified), resit 18-19*

(a) Consider a modification of the DFA model, called NEFA. The NEFA model is identical to the DFA with the extra requirement that $q_0 \in F$. Is the NEFA model equally expressive as the DFA model? If so, explain how we can construct an NEFA from a DFA. If not, give an example of a DFA that has no NEFA equivalent and argue why (no full proof is needed).

> **Solution:** Take a machine D with $L(D) = \emptyset$. We cannot make an NEFA with this language since every NEFA will accept ε, and thus the language of the NEFA cannot be empty.

(b) Take two DFAs D_1 and D_2. Now take the language $L = (L(D_1) \cap L(D_2))^*$. Is L regular? Explain your answer in at most 5 lines.

> **Solution:** Yes, regularity is closed under both $\cap$ and $*$.

See next page for an optional extra ~~exercise~~ puzzle.

Optional extra exercise This last exercise is not part of the learning material. However, if you like puzzles and have some time left, you might like to do this challenging puzzle.

In one of your adventures, you try to steal some treasure from an old mystical mansion. As this mansion has no door, you have build a teleportation device to teleport inside. This works as you expected and you find the treasure you were looking for. However, as the mansion still has no door, it will be a challenge to get out. The room you are in is mostly empty. The only things in there are 2 teleportation devices at the other side of the room. As you have no other choice, you decide to try one of them, and arbitrarily choose the left one. You teleport to an identical room, again with 2 teleportation devices. Now, you decide to use the right teleportation device. Again, you end up in an identical room. Since it's identical, you don't know whether this is a room you haven't been in before, the room you started in, or even the room you just left. You take the left teleportation device ending up in an identical room, and then the right teleportation device again, with the same result. Starting to lose hope, you enter the right teleportation device, ending up in a different room. This room has white walls, in contrast to the grey walls of before. This gives you hope. If there are different rooms, it might be easier to deduce where you are. Given the size of the mansion and the rooms, you know there are at most 4 rooms in the mansion. Also, you know that there is at least 1 teleportation device that leads outside. You use a few more teleportation devices (given below), but then the walls start to crumble. The mansion is falling apart. You have to get outside within 4 uses of a teleportation device. You can't be sure if it is possible, but which 4 teleportation devices should you use to have the only chance to get outside?

Start	Left	Right	Left	Right	Right	Left	Left	Left	Left	Left	Left
Grey	Grey	Grey	Grey	Grey	White	Grey	Grey	Grey	Grey	Grey	Grey

Left	Right	Right	Right	Left	Right	Left	?	?	?	?
Grey	Grey	White	White	Grey	White	Grey				Outside

- Each room has 2 teleportation devices, a left one and a right one
- Teleportation devices are deterministic (the same device in the same room will always lead to the same room)
- At least 1 teleportation device leads outside
- There are at most 4 rooms
- Rooms will not change colour (the same room will always have the same colour)
- After you have taken the teleportation devices as given in the table (and entered rooms with wall colours as given in the table), which 4 teleportation devices should you take to have a chance to get outside?

Solution: $Right \to Right \to Right \to Right$

- Extra: The last sentence of the story says "You can't be sure if it is possible". Why?

Solution: There are multiple room configurations which give the output in the table. In some of them, the teleportation device that leads outside is in a room you can't get to, because no teleportation device goes there (such a room corresponds to an unreachable state). In those cases, it's not possible to get outside. Note that there are still multiple possible room configurations that do have a solution, but 4 times right will work in all of them.

End of the assignment